BLUETOOTH ENABLED ROBOT CAR PRINTED CIRCUIT BOARD(PCB) DESIGN

STM32WB55 MICROCONTROLLER(BLUETOOTH ENABLED) AND TB6612FNG MOTOR DRIVER

BY

OYETORO OLUWAYOMI DAVID

NITDA IT HUB (NitHub)

UNIVERSITY OF LAGOS

AKOKA, NIGERIA

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1.0 Background Study

The STM32WB55 is a versatile and widely-used 32-bit microcontroller for embedded development, offering a balance of performance, features, and cost-effectiveness. It offers high performance and low power consumption, making it suitable for a wide range of embedded applications. The "WB" in STM32WB55 indicates its support for Bluetooth Low Energy (BLE) and IEEE 802.15.4 wireless protocols, making it suitable for various wireless connectivity applications. This capability is particularly useful in IoT devices and applications requiring short-range wireless communication. The STM32WB55 microcontroller can be programmed in STM32CubeIDE and C/C++ using libraries provided by STMicroelectronics or third-party libraries.
Due to its performance, peripherals, and low-power features, the STM32WB55 microcontroller is commonly used in a wide range of embedded applications including industrial control systems, consumer electronics, smart appliances, medical devices, automotive systems, and more.

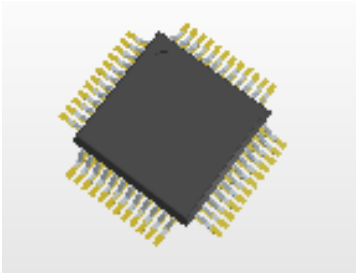
The TB6612FNG is a dual motor driver IC (Integrated Circuit) that is commonly used in robotics and other projects requiring motor control. It can control two DC motors. This dual functionality makes it versatile for various applications, including wheeled robots, robotic arms, and more. It operates at a wide range of voltages, typically from 2.7V to 5.5V. The IC incorporates current limit functionality, which protects the motors from damage due to overcurrent conditions. This feature ensures the longevity of both the motor and the driver IC itself. The TB6612FNG accepts PWM inputs for controlling the speed and direction of the motors it drives. It has a straightforward control interface, typically requiring only a few control pins to operate which makes it easy to integrate into various projects and microcontroller platforms. The TB6612FNG is available in compact packages, such as SSOP and QFN, which helps save space on PCBs (Printed Circuit Boards) and makes it suitable for applications with size constraints. Compared to more advanced motor drivers, the TB6612FNG provides a cost-effective solution for basic motor control needs without compromising on essential features.

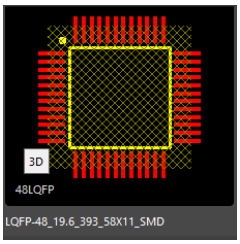
2.0 Aim and Objectives

To design a Printed circuit Board for a robot car using STM32WB55 Microcontroller(Bluetooth enabled) and TB6612FNG motor driver using Altium designer

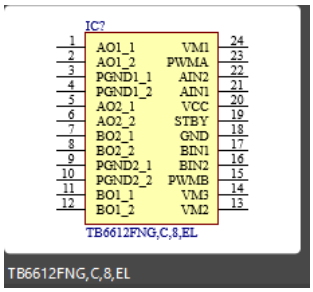
3.0 Apparatus/Component Used

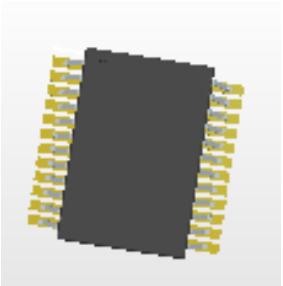
1. STM32WB55CEU6 Microcontroller

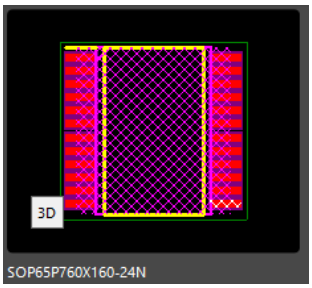


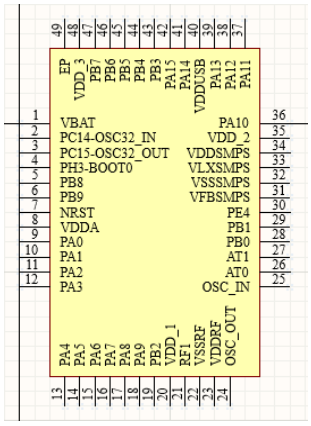


2. TB6612FNG Motor Driver IC

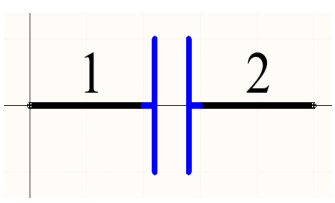


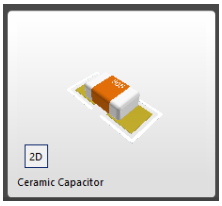


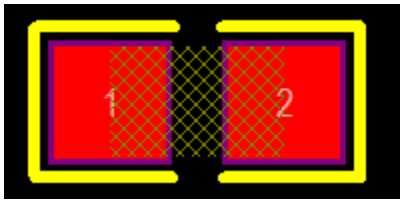




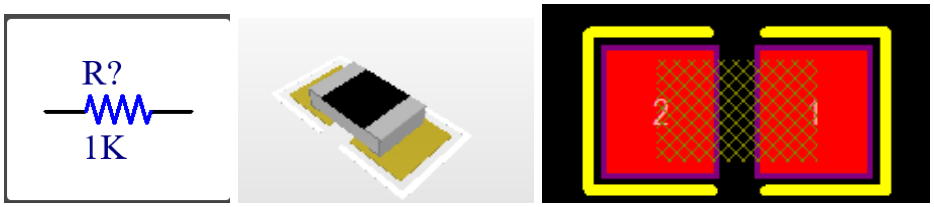
3. Capacitors (4.7uF,100nF,10pF,0.8pF,0.3pF,10uF,0.1uF,)



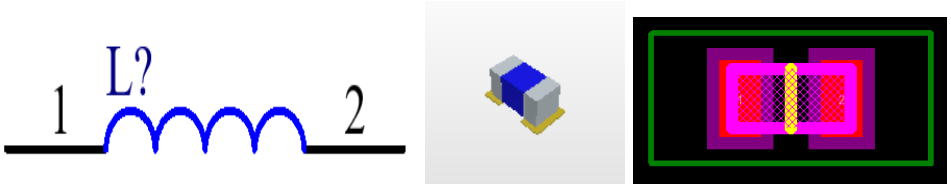




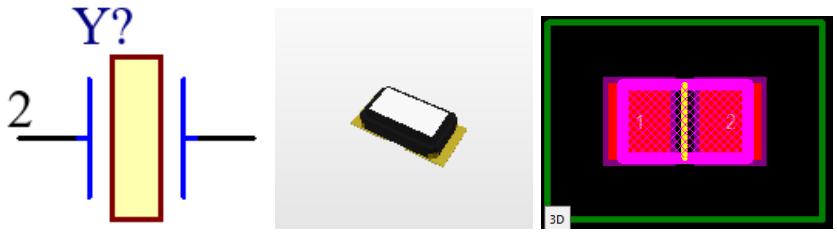
4. Resistors (100 ohms,220 ohms,5.1k ohms)

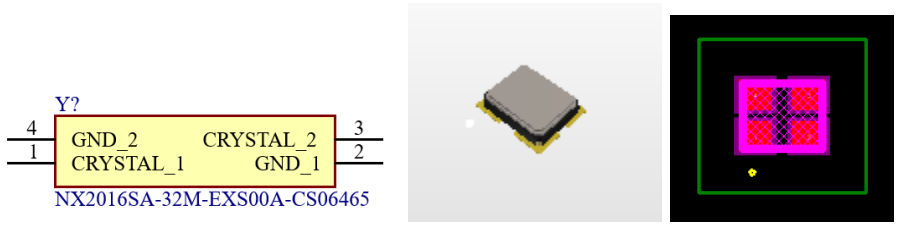


5. Inductors(10nH,10uH,2.7nH)



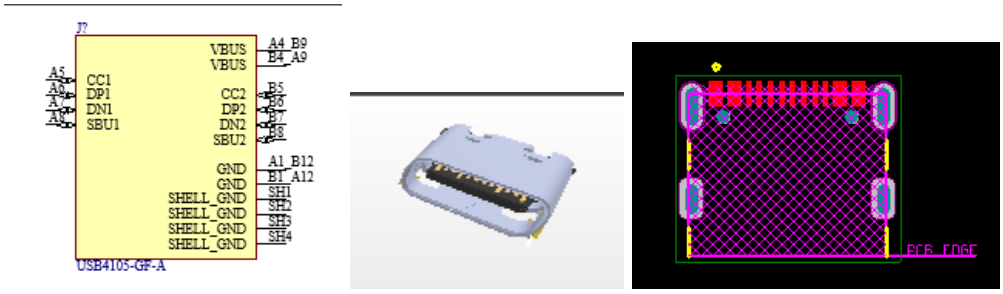
6. Crystal Oscillators (32.768KHz,32Mhz)



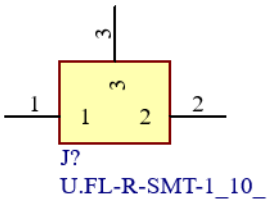


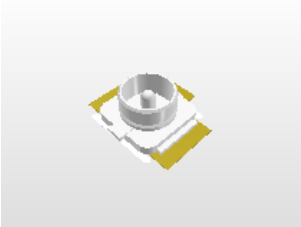
7. Connectors

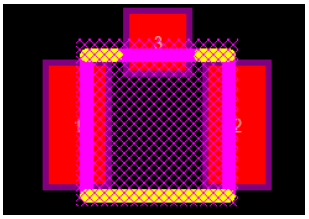
USB C



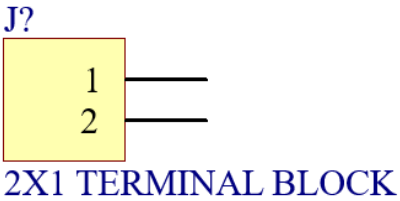
Antenna Connector

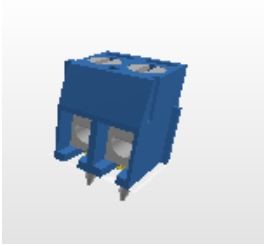


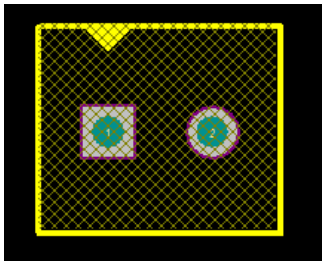




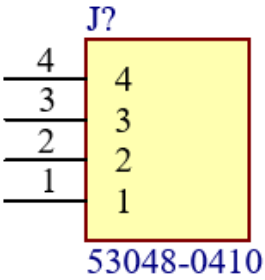
2x1 Connector

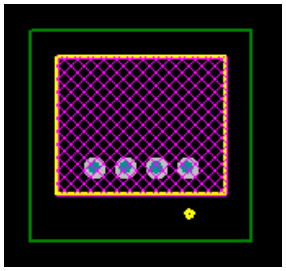




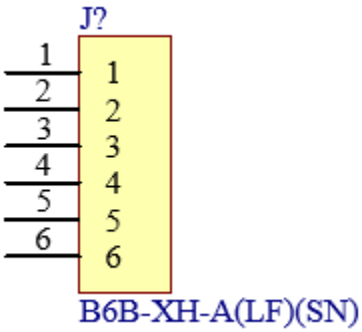


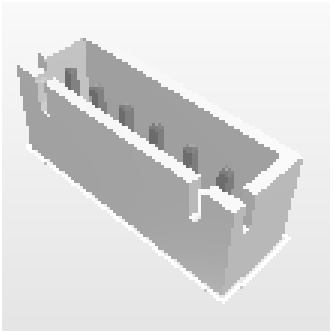
4x1 Connector

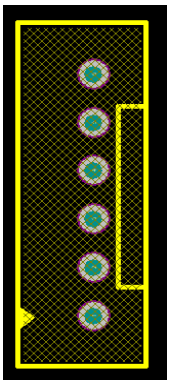




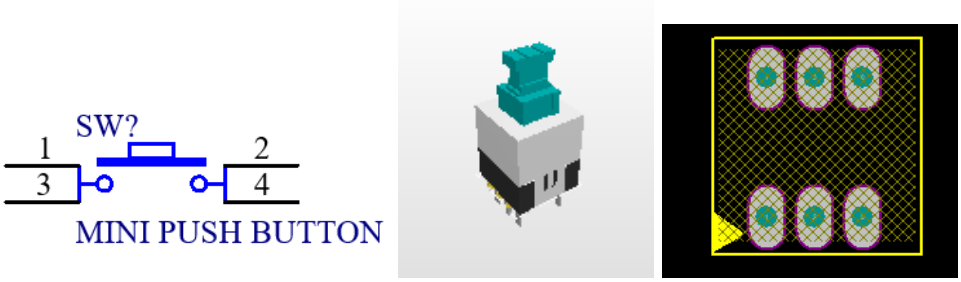
6x1 Connector





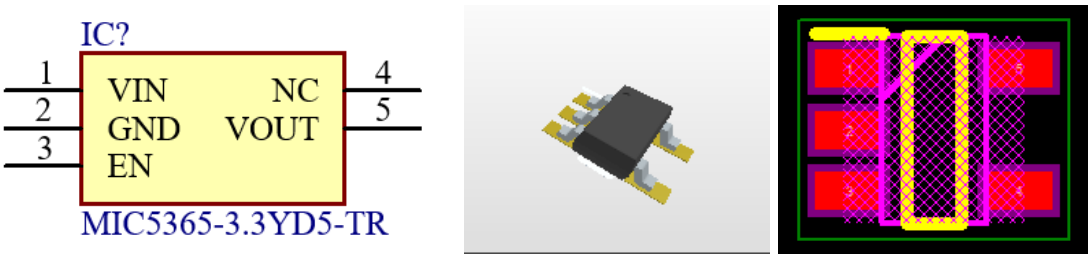


8. Switch



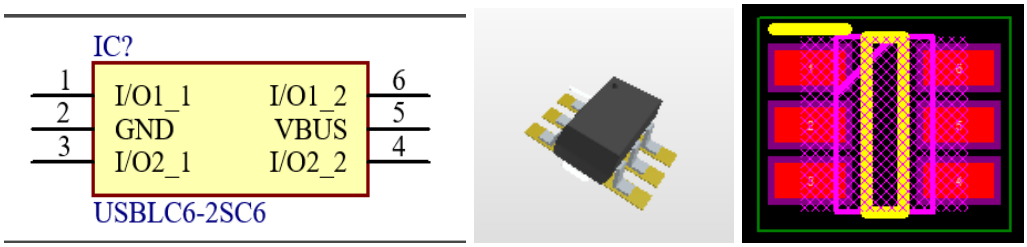
9. LED (Red)

10. Regulator (5v-3.3v)



11. Low Pass Filter

12. USB Protector IC

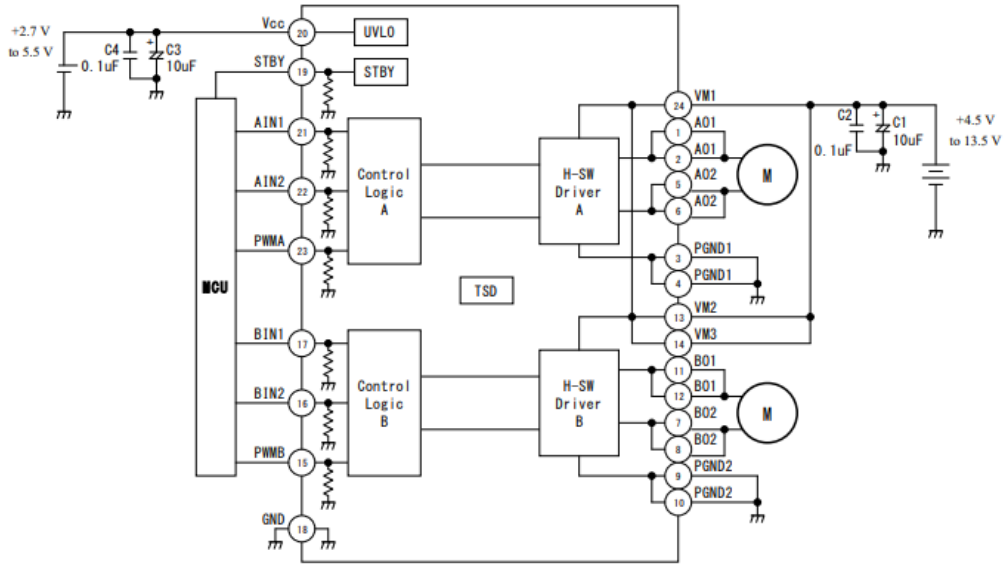


4.0 METHODOLOGY/ PROCEDURE

1. Create the components (symbol and footprint) in a library or download a library with the components needed and compile the library.
2. Create a new project, add new schematic and PCB document
3. On the schematic page, place the components and connect according to the circuit diagram for STM32F103 and TB6612FNG
4. Annotate the components on the schematic page
5. Import the schematic to the PCB document
6. Resize the PCB to the appropriate size
7. Place the components properly on the PCB board
8. Make the connection using wires(track) and vias on the PCB board
9. For the ground connection,do a polygon pour
10. Prepare the job output file to be sent company for the PCB printing (and assembly)

5.0 Diagrams

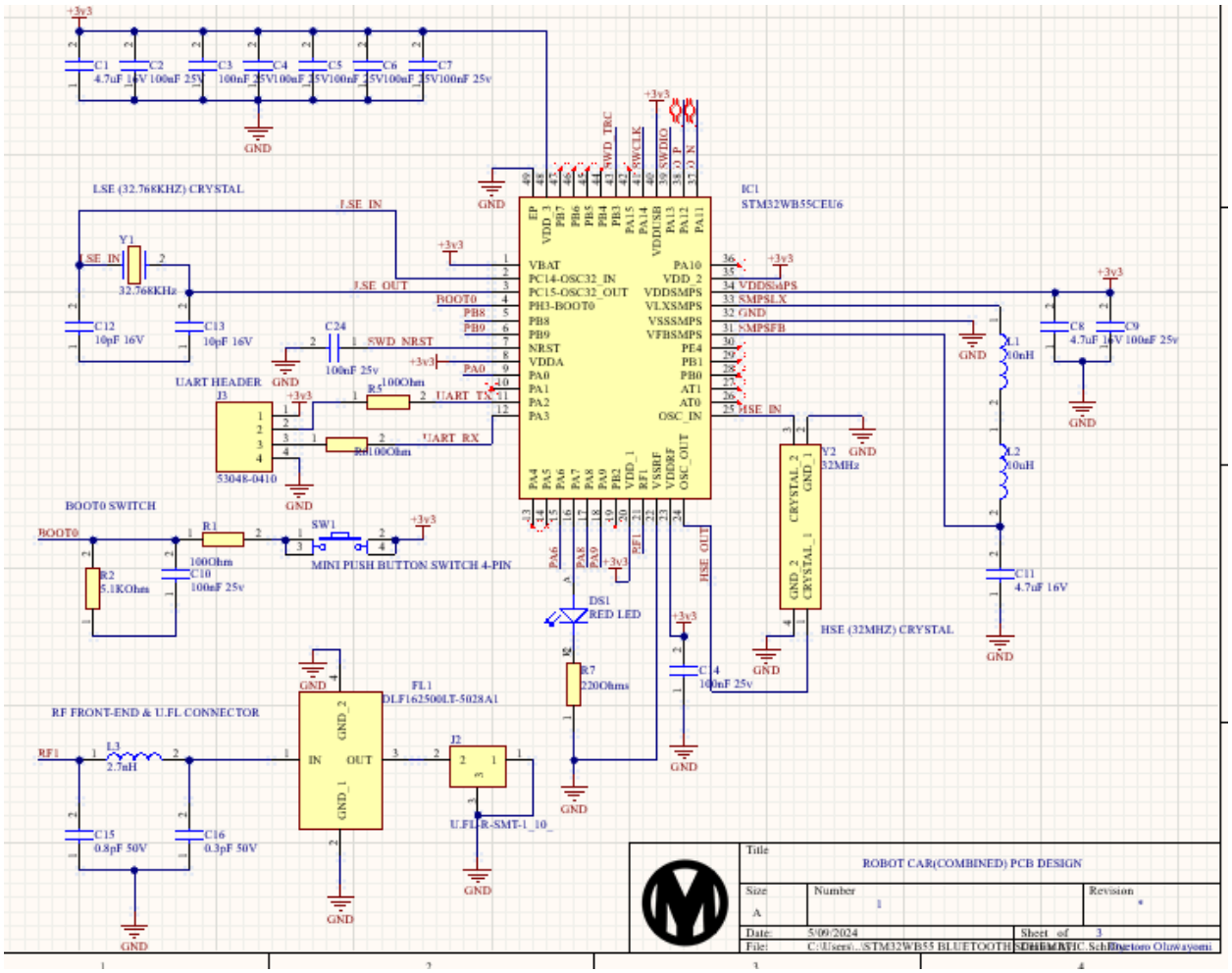
5.1 Circuit Diagram/ Pin out

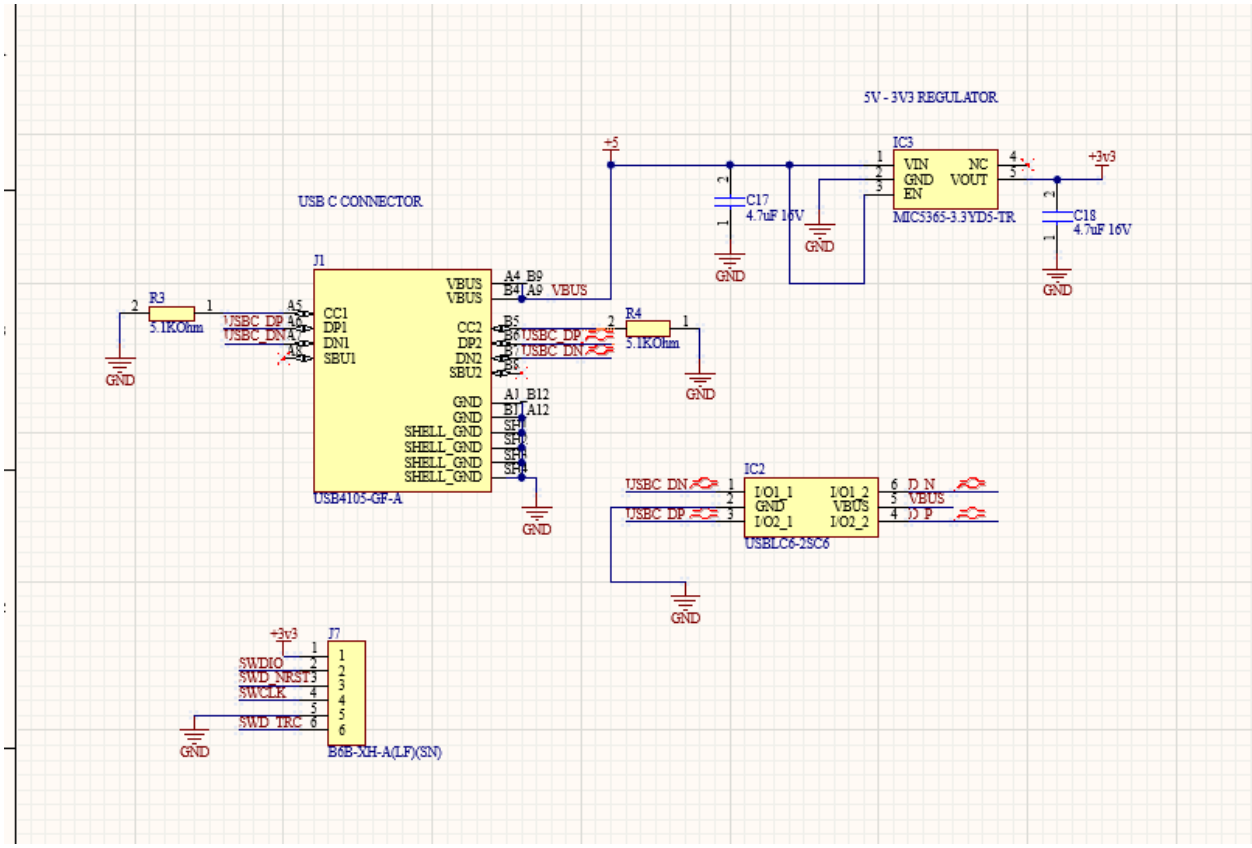


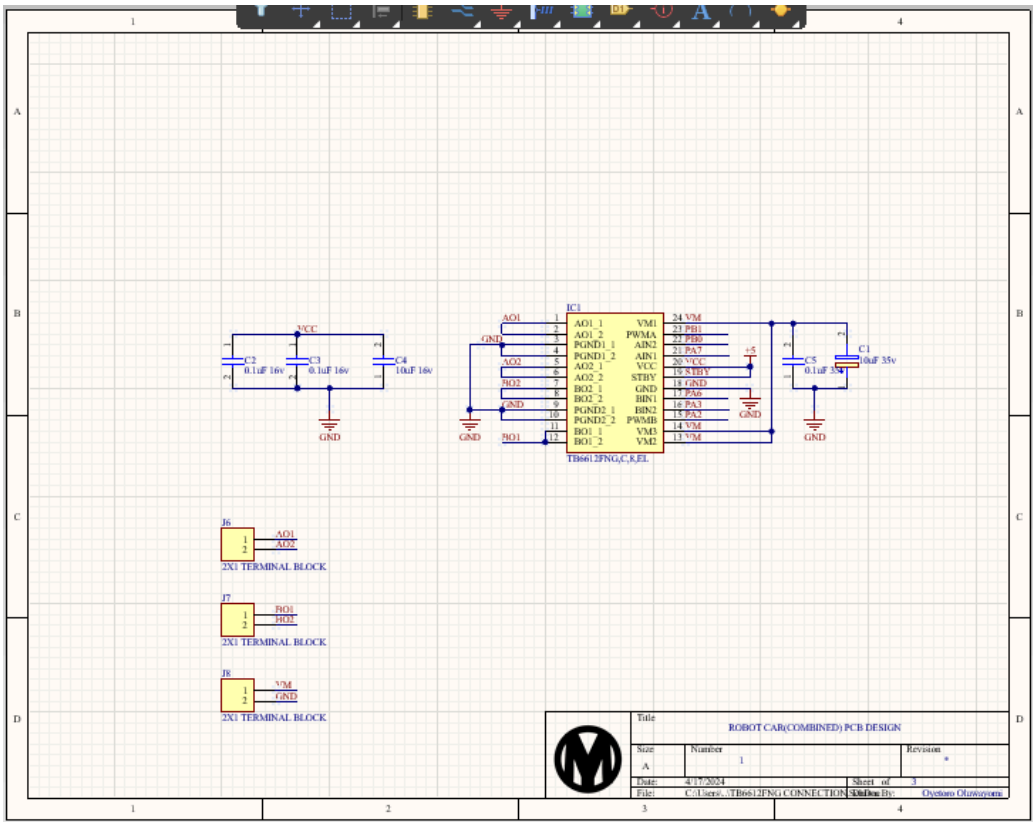
5.2 PCB board size

The size of the PCB is 58mm by 25.4mm.

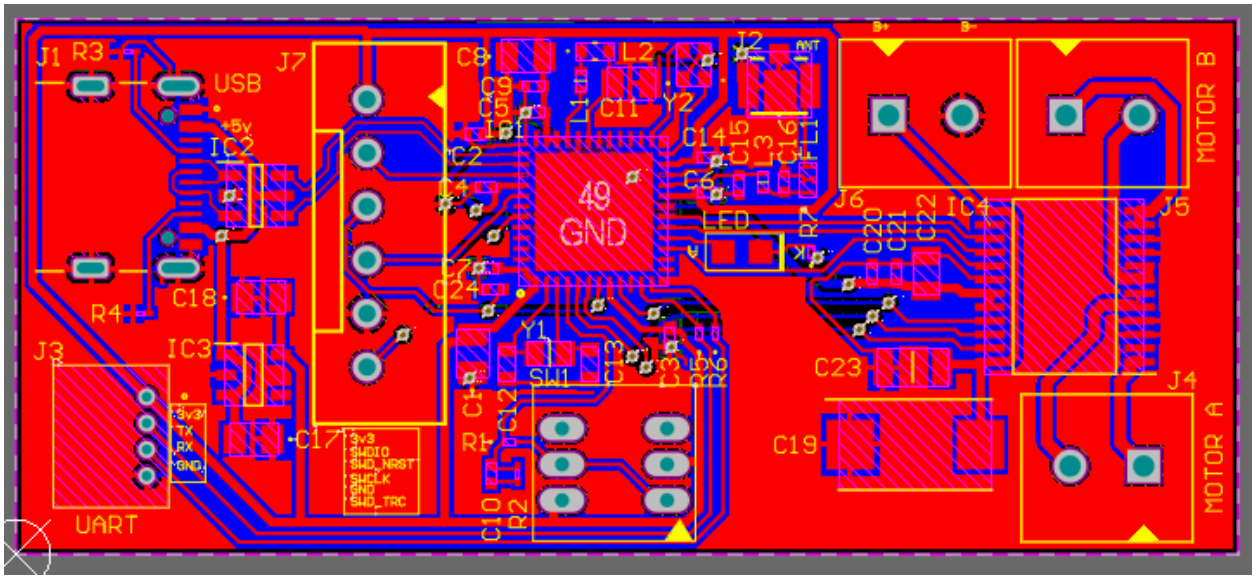
5.3 Schematic Diagram



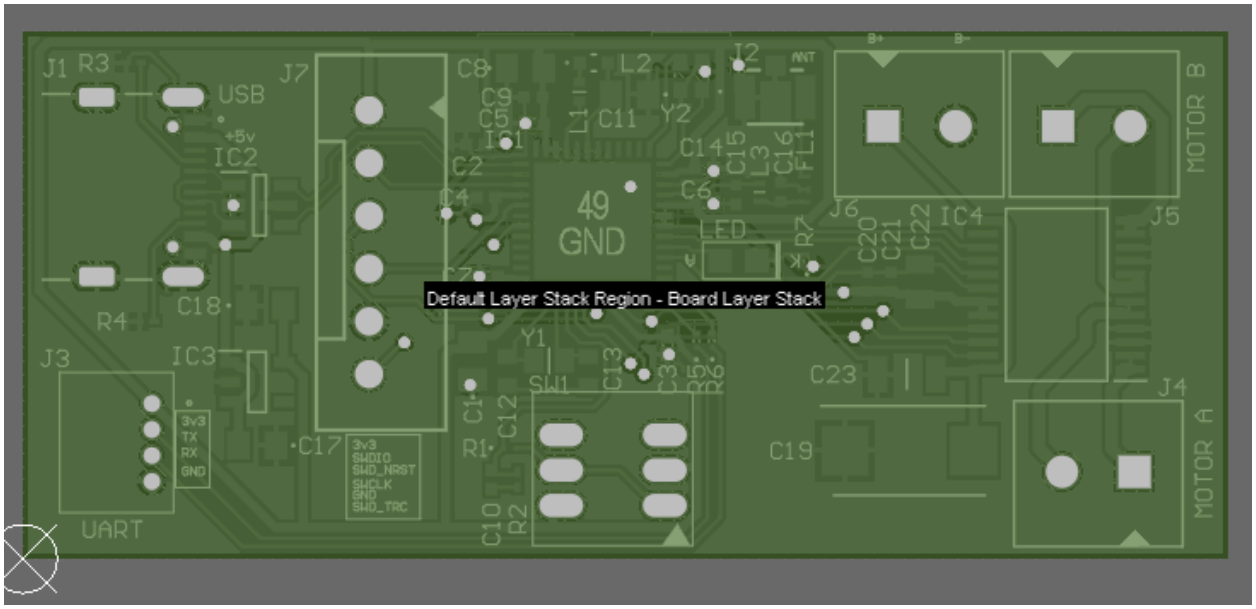

5V - 3V3 REGULATOR
USB C CONNECTOR
IC3
MIC5365-3.3YD5-TR
C17
4.7uF 16V
C18
4.7uF 16V
+5
+3v3
J1
USB4105-GF-A
R3
5.1KOhm
R4
5.1KOhm
USBC_DP
USBC_DN
VBUS
GND
IC2
USBLC6-2SC6
J7
B6B-XH-A(LF)(SN)
SWDIO
SWD_NRST
SWCLK
SWD_TRC


IC1
TB6612FNG,C,K,EL
C2
0.1uF 16v
C3
0.1uF 16v
C4
10uF 16v
C5
0.1uF 35v
C1
10uF 35v
VCC
GND
+5
J6
2X1 TERMINAL BLOCK
J7
2X1 TERMINAL BLOCK
J8
2X1 TERMINAL BLOCK
Title
ROBOT CAR(COMBINED) PCB DESIGN
Size
A
Number
1
Revision
Date: 4/17/2024
Sheet of 3
File: C:\Users...\TB6612FNG CONNECTION.SchDoc
Drawn By: Oyetaro Oluwayomi

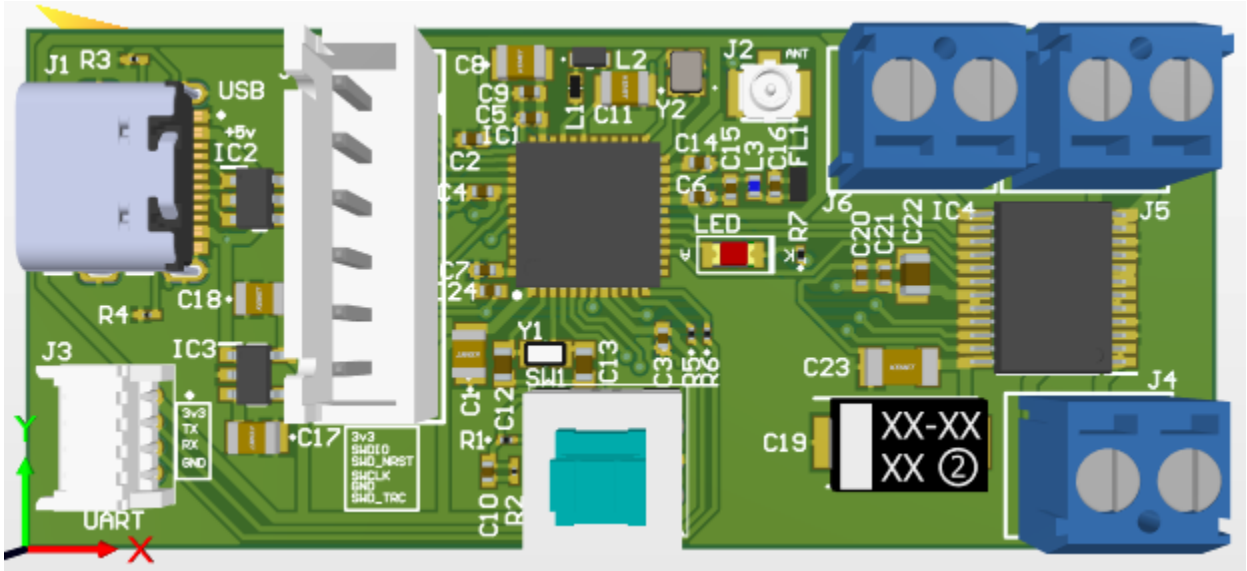
5.4 2-D PCB Board



Layer stack



6.0 3D PCB BOARD



7.0 CONCLUSION

The Printed Circuit board(combined) consisting of the STM32WB55 microcontroller(with bluetooth) and TB6612FNG motor driver for the robot car was successfully designed and ready for printing.